

```
>> dynare mcmc
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 22 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
6 block(s) found:
  4 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 15 equation(s)
    and 15 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 15 equation(s)
    and 15 feedback variable(s).

Preprocessing completed.
Preprocessing time: 0h00m00s.
```

STEADY-STATE RESULTS:

q	32.8507
k	4.41224
R	1.01351
b	143.013
x	0.0406367
varphi	0.0303425
mu	0.00686924
kp	2.20776
chi	1.02177
nu	0.00432944
eta	1.28813
zeta	1.02177
N	41.8193
phi	3.41978
Ne	41.5333
Nn	0.286025
xp	6.22046
bp	-101.193
Y	6.2611
eb	0
N_obs	0
Y_obs	0

EIGENVALUES:

Modulus	Real	Imaginary
8.246e-21	-8.246e-21	0
1.674e-16	-1.674e-16	0
2.636e-16	-2.636e-16	0
1.46e-09	1.46e-09	0
1.46e-09	-1.46e-09	0
0.1549	-0.009895	0.1545

0.1549	-0.009895	-0.1545
0.971	0.971	0
1.011	1.011	0
1.02	1.02	0
1.022	1.022	0
1.039	1.039	0
9.867e+17	9.867e+17	0
1.191e+34	-1.191e+34	0
Inf	-Inf	0

There are 7 eigenvalue(s) larger than 1 in modulus for 7 forward-looking variable(s)
The rank condition is verified.

===== Identification Analysis =====

Testing prior mean

Note that differences in the criteria could be due to numerical settings,
numerical errors or the method used to find problematic parameter sets.

Settings:

Derivation mode for Jacobians:	Analytic using sylvester equations
Method to find problematic parameters:	Nullspace and multicorrelation

coefficients

Normalize Jacobians:	Yes
Tolerance level for rank computations:	robust
Tolerance level for selecting nonzero columns:	1e-08
Tolerance level for selecting nonzero singular values:	1e-03

REDUCED-FORM:

All parameters are identified in the Jacobian of steady state and reduced-form solution matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):

All parameters are identified in the Jacobian of steady state and minimal system (rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):

All parameters are identified in the Jacobian of mean and spectrum (rank(Gbar) is full with tol = robust).

MOMENTS (ISKREV, 2010):

All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol = robust).

==== Identification analysis completed ====

You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): -1653.4871

f at the beginning of new iteration, 1653.4871445690

Predicted improvement: 6741319.082064427

lambda =	1;	f =	2829.3598491
lambda =	0.33333;	f =	1750.1851362
lambda =	0.11111;	f =	1656.3129666
lambda =	0.037037;	f =	72.6532307

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lambda = 0.012346; f = -24.6878525
lambda = 0.0041152; f = -83.0286713
lambda = 0.0013717; f = -79.2862856
lambda = 0.00045725; f = 96.4034062
lambda = 0.00015242; f = 578.4637395
Norm of dx 36.719
----
Improvement on iteration 1 = 1736.515815861
-----
f at the beginning of new iteration, -83.0286712924
Predicted improvement: 99.630970941
lambda = 1; f = 6.0061205
lambda = 0.33333; f = -43.0995646
lambda = 0.11111; f = -74.1666797
lambda = 0.037037; f = -84.2114941
lambda = 0.012346; f = -84.5873461
Norm of dx 0.14115
----
Improvement on iteration 2 = 1.558674846
-----
f at the beginning of new iteration, -84.5873461386
Predicted improvement: 0.223353203
lambda = 1; f = -84.8965958
Norm of dx 0.0022814
----
Improvement on iteration 3 = 0.309249646
-----
f at the beginning of new iteration, -84.8965957848
Predicted improvement: 0.100766950
lambda = 1; f = -85.0447134
lambda = 1.9332; f = -85.0936524
Norm of dx 0.0038124
----
Improvement on iteration 4 = 0.197056616
-----
f at the beginning of new iteration, -85.0936524006
Predicted improvement: 0.062043202
lambda = 1; f = -85.2172075
lambda = 1.9332; f = -85.3315510
lambda = 3.7372; f = -85.5499455
lambda = 7.2247; f = -85.9620006
lambda = 13.967; f = -86.7189361
lambda = 27; f = -88.0201216
lambda = 52.196; f = -89.8050161
lambda = 100.9; f = -89.0782154
Norm of dx 0.003997
----
Improvement on iteration 5 = 4.711363728
-----
f at the beginning of new iteration, -89.8050161286
Predicted improvement: 4.993709843
lambda = 1; f = -95.3044573
Norm of dx 0.3703
----
Improvement on iteration 6 = 5.499441130
-----
f at the beginning of new iteration, -95.3044572586
Predicted improvement: 7.511734010

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lambda =          1; f =          -95.3044537
lambda =    0.33333; f =          -99.0732451
lambda =    0.64439; f =          -85.7874457
lambda =    0.4339; f =          -98.6366276
Norm of dx    0.12224
----
Improvement on iteration 7 =          3.768787868
-----
f at the beginning of new iteration,          -99.0732451263
Predicted improvement:          5.127026974
lambda =          1; f =          -96.2922508
lambda =    0.33333; f =          -100.2156569
Norm of dx    0.26767
----
Improvement on iteration 8 =          1.142411777
-----
f at the beginning of new iteration,          -100.2156569038
Predicted improvement:          0.153196257
lambda =          1; f =          -100.3617589
Norm of dx    0.019541
----
Improvement on iteration 9 =          0.146101974
-----
f at the beginning of new iteration,          -100.3617588776
Predicted improvement:          0.153310548
lambda =          1; f =          -100.6091364
lambda =    1.9332; f =          -100.7253146
Norm of dx    0.01386
----
Improvement on iteration 10 =          0.363555702
-----
f at the beginning of new iteration,          -100.7253145794
Predicted improvement:          0.251959409
lambda =          1; f =          -101.2134721
lambda =    1.9332; f =          -101.6400542
lambda =    3.7372; f =          -102.3823486
lambda =    7.2247; f =          -103.4874513
lambda =   13.967; f =          -104.1848073
Norm of dx    0.0064223
----
Improvement on iteration 11 =          3.459492694
-----
f at the beginning of new iteration,          -104.1848072733
Predicted improvement:          5.274999019
lambda =          1; f =          -114.2316231
lambda =    1.9332; f =          -121.5543962
lambda =    3.7372; f =          48.8958043
lambda =    2.5164; f =          -121.1325084
Norm of dx    0.053525
----
Improvement on iteration 12 =          17.369588880
-----
f at the beginning of new iteration,          -121.5543961534
Predicted improvement:          5.754476418
lambda =          1; f =          2.5157242
lambda =    0.33333; f =          -122.3555760
lambda =    0.11111; f =          -122.5887421
lambda =    0.2148; f =          -122.9725699

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Norm of dx    0.080216
-----
Improvement on iteration 13 =          1.418173700
-----
f at the beginning of new iteration,      -122.9725698531
Predicted improvement:          4.577120226
lambda =          1; f =          -127.9099922
Norm of dx    0.090964
-----
Improvement on iteration 14 =          4.937422359
-----
f at the beginning of new iteration,      -127.9099922122
Predicted improvement:          1.650117936
lambda =          1; f =          -130.1717229
Norm of dx    0.021079
-----
Improvement on iteration 15 =          2.261730687
-----
f at the beginning of new iteration,      -130.1717228990
Predicted improvement:          0.498711025
lambda =          1; f =          -130.5672820
Norm of dx    0.043039
-----
Improvement on iteration 16 =          0.395559118
-----
f at the beginning of new iteration,      -130.5672820165
Predicted improvement:          0.276062205
lambda =          1; f =          -130.7404851
Norm of dx    0.011928
-----
Improvement on iteration 17 =          0.173203091
-----
f at the beginning of new iteration,      -130.7404851077
Predicted improvement:          0.031803732
lambda =          1; f =          -130.7687845
Norm of dx    0.004795
-----
Improvement on iteration 18 =          0.028299418
-----
f at the beginning of new iteration,      -130.7687845252
Predicted improvement:          0.002049654
lambda =          1; f =          -130.7704737
Norm of dx    0.0012485
-----
Improvement on iteration 19 =          0.001689186
-----
f at the beginning of new iteration,      -130.7704737112
Predicted improvement:          0.000064121
lambda =          1; f =          -130.7705439
Norm of dx    8.397e-05
-----
Improvement on iteration 20 =          0.000070151
-----
f at the beginning of new iteration,      -130.7705438625
Predicted improvement:          0.000001876
lambda =          1; f =          -130.7705481
lambda =          1.9332; f =          -130.7705517
lambda =          3.7372; f =          -130.7705575

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lambda =      7.2247; f =      -130.7705648
lambda =     13.967; f =      -130.7705639
Norm of dx 3.4682e-05
----
Improvement on iteration 21 =      0.000020898
-----
f at the beginning of new iteration,      -130.7705647609
Predicted improvement:      0.000017945
lambda =      1; f =      -130.7706009
lambda =     1.9332; f =      -130.7706335
lambda =     3.7372; f =      -130.7706940
lambda =     7.2247; f =      -130.7708005
lambda =    13.967; f =      -130.7709682
lambda =      27; f =      -130.7711491
Norm of dx 0.00022667
----
Improvement on iteration 22 =      0.000584380
-----
f at the beginning of new iteration,      -130.7711491409
Predicted improvement:      0.000500147
lambda =      1; f =      -130.7720681
lambda =     1.9332; f =      -130.7727765
lambda =     3.7372; f =      -130.7737378
Norm of dx 0.0064468
----
Improvement on iteration 23 =      0.002588643
-----
f at the beginning of new iteration,      -130.7737377834
Predicted improvement:      0.000850051
lambda =      1; f =      -130.7746023
Norm of dx 0.015413
----
Improvement on iteration 24 =      0.000864498
-----
f at the beginning of new iteration,      -130.7746022816
Predicted improvement:      0.000009037
lambda =      1; f =      -130.7746080
Norm of dx 0.00036341
----
Improvement on iteration 25 =      0.000005754
-----
f at the beginning of new iteration,      -130.7746080353
Predicted improvement:      0.000000069
lambda =      1; f =      -130.7746079
lambda =     0.33333; f =      -130.7746080
lambda =     0.11111; f =      -130.7746080
lambda =     0.037037; f =      -130.7746080
lambda =     0.012346; f =      -130.7746080
lambda =     0.0041152; f =      -130.7746080
lambda =     0.0013717; f =      -130.7746080
lambda =     0.00045725; f =      -130.7746080
lambda =     0.00015242; f =      -130.7746080
lambda = 5.0805e-05; f =      -130.7746080
lambda = 1.6935e-05; f =      -130.7746080
lambda = 5.645e-06; f =      -130.7746080
lambda = 1.8817e-06; f =      -130.7746080
lambda = 6.2723e-07; f =      -130.7746080
lambda = 2.0908e-07; f =      -130.7746080

```

```
lambda = 6.9692e-08; f = -130.7746080
lambda = 2.3231e-08; f = -130.7746080
lambda = 7.7435e-09; f = -130.7746080
lambda = 2.5812e-09; f = -130.7746080
```

```
lambda =
```

```
-6.2723e-07
```

```
lambda = -6.2723e-07; f = -130.7746080
lambda = -2.0908e-07; f = -130.7746080
lambda = -6.9692e-08; f = -130.7746080
lambda = -2.3231e-08; f = -130.7746080
lambda = -7.7435e-09; f = -130.7746080
lambda = -2.5812e-09; f = -130.7746080
Norm of dx 0.00016944
```

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```

```
Improvement on iteration 26 = 0.000000000
improvement < crit termination
smallest step still improving too slow, reversed gradient
```

```
Final value of minus the log posterior (or likelihood):-130.774608
```

```
MODE CHECK
```

```
Fval obtained by the optimization routine: -130.774608
```

```
RESULTS FROM POSTERIOR ESTIMATION
```

```
parameters
```

	prior mean	mode	s.d.	prior	pstdev
gamma_q	0.0000	0.5175	0.4825	norm	0.5000
gamma_N	0.0000	1.0697	0.1449	norm	0.5000

```
standard deviation of shocks
```

	prior mean	mode	s.d.	prior	pstdev
eN	0.0100	0.0067	0.0011	inv	0.0050
eY	0.0100	0.0467	0.0046	inv	0.0050

```
Log data density [Laplace approximation] is 117.864928.
```

```
Estimation::mcmc: Multiple chains mode.
```

```
Estimation::mcmc: Searching for initial values...
```

```
Estimation::mcmc: Initial values found!
```

```
Estimation::mcmc: Write details about the MCMC... Ok!
```

```
Estimation::mcmc: Details about the MCMC are available in mcmc/metropolis\mcmc_mh_history_0.mat
```

```
Estimation::mcmc: Number of mh files: 1 per block.
```

```
Estimation::mcmc: Total number of generated files: 2.
```

```
Estimation::mcmc: Total number of iterations: 125000.
```

```
Estimation::mcmc: Current acceptance ratio per chain:
```

```
Chain 1: 47.6144%
```

```
Chain 2: 48.0768%
```

```
Estimation::mcmc: Total number of MH draws per chain: 125000.
```

Estimation::mcmc: Total number of generated MH files: 1.
 Estimation::mcmc: I'll use mh-files 1 to 1.
 Estimation::mcmc: In MH-file number 1 I'll start at line 25001.
 Estimation::mcmc: Finally I keep 100000 draws per chain.

MCMC Inefficiency factors per block

Parameter	Block 1	Block 2
SE_eN	22.218	23.850
SE_eY	20.320	19.250
gamma_q	19.784	20.528
gamma_N	18.977	20.274

Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 1.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.007	0.001	0.159	0.760	
0.757	0.770				
SE_eY	0.048	0.005	0.000	0.163	
0.117	0.112				
gamma_q	0.474	0.478	0.000	0.327	
0.302	0.187				
gamma_N	1.066	0.144	0.000	0.387	
0.371	0.259				

Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 2.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.007	0.001	0.021	0.587	
0.552	0.534				
SE_eY	0.049	0.005	0.000	0.353	
0.311	0.258				
gamma_q	0.477	0.488	0.002	0.457	
0.432	0.363				
gamma_N	1.064	0.145	0.082	0.647	
0.604	0.527				

Univariate convergence diagnostic, Brooks and Gelman (1998):

Parameter 1... Done!
 Parameter 2... Done!
 Parameter 3... Done!
 Parameter 4... Done!

marginal density: I'm computing the posterior mean and covariance... Done!
marginal density: I'm computing the posterior log marginal density (modified harmonic mean)...
Done!

ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 117.943941.

parameters

	prior mean	post. mean	90% HPD interval		prior	pstdev
gamma_q	0.000	0.4749	-0.3275	1.2571	norm	0.5000
gamma_N	0.000	1.0653	0.8299	1.3020	norm	0.5000

standard deviation of shocks

	prior mean	post. mean	90% HPD interval		prior	pstdev
eN	0.010	0.0072	0.0054	0.0091	invg	0.0050
eY	0.010	0.0485	0.0404	0.0562	invg	0.0050

--- Parameters Post. Std ---

gamma_q: 0.482568

gamma_N: 0.144403

--- Shocks Post. Std ---

eN: 0.001166

eY: 0.004953

Total computing time : 0h21m25s

>>